Session 31 (AIML) - Assignment Questions

Q1. (HuggingFace via API)

- Install the required packages using uv: langchain-huggingface, transformers, and torch.
- Create a .env file and store your HUGGINGFACEHUB_API_TOKEN.
- Write a Python program that uses HuggingFaceEndpoint with the model deepseek-ai/DeepSeek-R1.
- Convert it into a chat model using ChatHuggingFace and ask it to introduce itself.
- Print the response.

Q2. (Using Local HuggingFace Model)

- Load a local model using HuggingFacePipeline with the model ID TinyLlama/TinyLlama-1.1B-Chat-v1.0.
- Set the task as "text-generation".
- Convert it into a chat model using ChatHuggingFace.
- Ask the model: "Introduce Yourself in 100 words".
- Print the complete response.

Q3. (Comparing API vs Local Model)
Write a program that asks the **same question** to both:

1. HuggingFace API model (deepseek-ai/DeepSeek-R1)
2. Local model (TinyLlama/TinyLlama-1.1B-Chat-v1.0)

Question: "What is the difference between AI and Machine Learning?"

Print both responses clearly labeled as “API Model Response” and “Local Model Response”. Write a short observation about the difference in quality or speed.

Q4. (Prompt Template with System Message)
Using any chat model (Groq or HuggingFace):

- Create a ChatPromptTemplate with:
  - A system message: "You are a very polite and helpful AI assistant."
  - A human message that takes user input.
- Take input from the user.
- Format the prompt and send it to the model.
- Print the response.

Q5. (Mini Project – Dual Mode Chatbot)
Build a simple command-line chatbot that:

1. Allows the user to choose between **API Mode** (HuggingFaceEndpoint) or **Local Mode** (HuggingFacePipeline).
2. Uses a polite system prompt via ChatPromptTemplate.
3. Continuously takes user input and gives responses.
4. Exits when the user types "exit".

Add proper comments and error handling (e.g., if the API token is missing).

Optional Task:

Create a frontend interface for your Agentic AI using Streamlit (or any other frontend technology such as Gradio, React, or HTML/CSS/JS).